

The Role of TiO_2 and ZnO Nanoparticles in Optimizing the UV Resistance, Thermal Stability, and Mechanical Properties of Polyethylene-Based Composite Materials

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Abstract. Polyethylene (PE) is susceptible to photo-oxidative degradation under ultraviolet (UV) irradiation, exhibits limited thermal/oxidative stability, and faces stiffness-toughness trade-offs. This work examines TiO_2 and ZnO nanoparticles-employed as bare and surface-modified fillers with a PE-g-MA compatibilizer-to optimize the UV resistance, thermal stability, and mechanical performance of PE composites. Composites were produced by melt compounding and solution casting. Accelerated UV weathering was performed, and specimens were characterized by FTIR carbonyl index, UV-Vis, TGA/DSC/DMA, and SEM/TEM; differences were evaluated using ANOVA. Coated TiO_2/ZnO at 1 – 3 wt% delivered the best balance, increasing oxidative-induction time, raising TGA onset temperature, and elevating crystallinity and storage modulus while preserving tensile strength after aging. Surface modification suppressed photocatalytic discoloration and embrittlement, whereas compatibilization improved interfacial stress transfer. The improvements arise from UV attenuation, heterogeneous nucleation, and strengthened polymer-filler interfaces. The findings demonstrate applicability to industrial packaging and outdoor parts; limitations include agglomeration and viscosity growth at higher loadings.

1 Introduction

Polyethylene (PE) is susceptible to photo-oxidative degradation under ultraviolet (UV) irradiation, where chromophore activation initiates radical oxidation, chain scission, and surface embrittlement, eroding strength and ductility [1-2]. Manifestations include carbonyl build-up, yellowing, and microcracking during service. Inorganic UV absorbers such as TiO_2

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and ZnO are investigated to attenuate radiation via absorption/scattering; however, photocatalytic activity can accelerate oxidation unless particle surfaces are passivated [3]. Mitigation strategies employ inorganic shells (SiO_2, Al_2O_3) and organic treatments (silanes, fatty acids) to limit charge transfer while preserving screening [4]. Accelerated weathering shows low nanoparticle loadings maintain mechanical and color properties versus neat PE in service when dispersion and interfaces are rigorously controlled. Yet standardized processing-structure-property guidelines are lacking: dispersion metrics, interface chemistry, and optimal loading/size remain unsettled. Compatibilization-e.g., polyethylene-g-maleic anhydride (PE-g-MA)-enhances interfacial bonding and reduces agglomeration [5]. This work tests whether (i) coated TiO_2/ZnO improve UV retention and thermal stability, (ii) PE-g-MA increases durability, and (iii) 1-3 wt% loading yields the best balance of UV protection, thermal stability, stiffness, and toughness under accelerated weathering. We combine melt compounding and solution casting with multiscale characterization to quantify links between surface chemistry, dispersion, and performance. Organic UV filters primarily absorb UV-B, while inorganic ZnO/TiO_2 scatter and attenuate UV-A but compromise transparency. Benzotriazole, benzophenone, cyanoacrylate, and triazines are prevalent; combining organic-inorganic agents afford broad-spectrum protection overall (Figure 1).

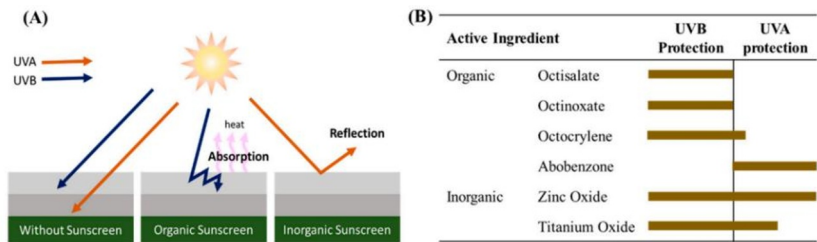


Fig. 1. (A) Schematic illustration of the way in which organic and inorganic sunscreens protect against ultraviolet (UV) radiation and (B) the active ingredients for protection against UV-A (320-400 nm) and UV-B (280-320 nm) rays [6].

2 Materials and Methods

2.1 Materials

Commercial high-density polyethylene (HDPE) and linear low-density polyethylene (LLDPE) grades were selected based on melt flow index (MFI) and weight-average molecular weight (Mw) appropriate for extrusion/press molding. Exact supplier names and lot numbers are recorded in the Supplementary Materials. Titanium dioxide (TiO_2 ; mixed anatase/rutile, primary size 15-50 nm) and zinc oxide (ZnO ; wurtzite, 20-60 nm) nanoparticles were used either uncoated or with inorganic shells (SiO_2 or Al_2O_3) and/or organic surface treatments (silanes such as APTES/KH-550, or stearic-acid capping) to mitigate photocatalysis and promote compatibility. A maleic-anhydride-grafted polyethylene compatibilizer (PE-g-MA) was employed at 1 – 3 wt% relative to polymer to enhance interfacial adhesion. All powders and pellets were oven-dried at 80-100 °C for 6-12 h prior to compounding to minimize moisture-induced defects.

2.2 Composite preparation

Three fabrication routes were employed to decouple dispersion from interfacial effects.

Melt compounding: A co-rotating twin-screw extruder (160 – 200 °C; 100 – 300 rpm; 2 – 4 min residence) produced a 20 wt% NP/PE masterbatch that was let-down to target loadings. Gravimetric feeding stabilized throughput; a vacuum vent removed volatiles. PE-g-MA was dosed with the neat polymer or preblended into the masterbatch. Strands were pelletized and compression-moulded into sheets. (b) Solution casting: PE was dissolved in refluxing xylene or decalin (≥ 110 °C); nanoparticles were probe-sonicated (20–60 min) in the same solvent and mixed with the solution. Films were cast, solvent evaporated, and porosity reduced by hot pressing slightly above T_m . (c) In-situ polymerization (optional): Nanoparticles were dispersed in monomer/catalyst media to promote molecular embedding.

Experimental design - Factors included NP type (TiO_2/ZnO), surface state (coated vs uncoated), loading (0, 0.5, 1, 2, 3, 5 wt%), and compatibilizer (0 vs 2 wt% PE-g-MA). Fractional factorial designs quantified main and interaction effects on UV, thermal, and mechanical responses. Thickness and specimen geometry complied with standards.

Specimen preparation - Sheets (≈ 2.0 – 3.2 mm) were compression-moulded via a preheat-press-cool schedule. Where required, injection-moulded bars were produced. Tensile dog-bones, flexural, and impact bars followed the cited specifications [7].

2.3 Characterization protocols

Morphology. Cryo-fractured cross-sections were sputter-coated with Au/Pd and examined by SEM to quantify filler dispersion, agglomerate size distributions, and interfacial integrity; a subset of specimens was microtomed/thinned for TEM to resolve primary particle networks. AFM phase and height maps furnished nanoscale roughness metrics and statistics of filler spatial distribution.

Structure and chemistry. Wide-angle X-ray diffraction ($2\theta = 5 - 80^\circ$, step 0.02°) identified crystalline phases of TiO_2/ZnO and tracked trends in polymer crystallinity. FTIR-ATR spectra ($4000 - 600\text{ cm}^{-1}$) were collected to follow oxidative changes; the carbonyl index (CI) was computed as the absorbance ratio A_{1715}/A_{1463} . Raman spectroscopy corroborated vibrational assignments and assessed pigment-related bands.

Thermal analysis. Thermogravimetric analysis (TGA/DTG) under N_2 (and, where indicated, in air) from ambient to 700 °C at $10\text{ }^\circ\text{C min}^{-1}$ yielded the onset temperature of degradation (T_{onset}), the peak rate temperature (T_{max}), and char residue. Differential scanning calorimetry (DSC) employed a heat-cool-heat protocol ($10\text{ }^\circ\text{C min}^{-1}$) to determine melting and crystallization temperatures (T_m, T_c) and enthalpies. The degree of crystallinity was calculated as

$$X_c = \frac{\Delta H_m}{\Delta H_m^0(1-\phi)} \quad (1)$$

where ϕ is the filler volume fraction. Oxidative induction time (OIT) was measured isothermally under flowing O_2 to quantify the effects of antioxidants and nanoparticles on oxidative stability.

Mechanical properties. Uniaxial tension followed ASTM D638 (Type I; crosshead rate per thickness) to obtain Young's modulus, yield strength, ultimate strength, and elongation at break [8]. Flexural performance was measured in three-point bending per ASTM D790. Notched impact resistance was determined using Izod/Charpy methods (ASTM D256). Dynamic mechanical analysis (DMA; 1 Hz, temperature ramp) provided E' , E'' , and $\tan \delta$ to characterize viscoelastic transitions [9].

UV durability and color. Accelerated weathering used fluorescent-UV exposure (ASTM G154; UVA-340 lamps) with a cyclic 8 h UV at 60 °C black-panel temperature followed by 4 h condensation at 50 °C; parallel xenon-arc tests adhered to ISO 4892-2. Exposures of 100-

500 h were applied, with CI, color, and mechanical testing after each interval. Yellowness Index (YI) was calculated per ASTM E313 [10]. Surface cracking was quantified by optical microscopy (areal crack density and mean crack length).

Rheology. Melt flow rate at 190 °C/2.16 kg followed ASTM D1238 to monitor processability changes due to fillers/compatibilizer. Small-amplitude oscillatory shear (190 °C, 0.1 – 100 rad s⁻¹) provided complex viscosity and viscoelastic moduli to assess network formation.

Thermal transport (optional). Through-thickness diffusivity/conductivity were determined by laser flash or transient plane source, with calibration and thickness corrections detailed in the Supplementary.

Surface energy and barrier (optional). Static contact angles (water/diiodomethane) yielded surface energy components; water-vapor transmission rate (WVTR) was measured on selected films to assess packaging relevance.

2.4 Characterization protocols

Mechanical, thermal, and optical data are reported as mean ± standard deviation with $n \geq 5$ for mechanical and $n \geq 3$ for thermal/optical tests. Normality and homoscedasticity were checked before one- or two-way ANOVA; pairwise comparisons used Tukey's HSD at $\alpha = 0.05$. Effect sizes and 95% confidence intervals are provided. Property retention after weathering was defined as the ratio of aged to unaged values. All instruments were calibrated per manufacturer procedures, and blanks/neat controls were run alongside filled formulations.

3 Results and Discussion

Dispersion–structure relationships. SEM/TEM revealed finely dispersed TiO_2 and ZnO domains with aggregate sizes largely below 100 nm in the best-performing formulations, whereas uncoated powders at ≥ 3 wt% tended to form larger secondary clusters. The addition of 2 wt% PE-g-MA systematically improved wetting at the particle–polymer interface, evidenced by cleaner particle outlines, fewer interfacial voids, and reduced pull-out features on cryo-fracture surfaces. These qualitative markers are consistent with stronger interfacial adhesion and more efficient load transfer under tensile loading, a prerequisite for concurrent gains in stiffness and property retention after weathering [11].

UV durability. Accelerated UV exposure led to the expected increases in carbonyl index (CI) and yellowness index (YI) for all samples, but the rate of change was significantly slower for nanofiller systems relative to neat PE. Coated TiO_2 /ZnO (SiO_2 or Al_2O_3 shells, or silane/stearate treatments) consistently outperformed uncoated analogues, showing lower CI build-up and diminished YI drift at equal dose, in line with the suppression of photocatalytic pathways on passivated oxide surfaces [11]. Mechanical retention (ultimate tensile strength $\sigma_{(b)}$, elongation at break $\epsilon_{(b)}$) tracked these chemical/color metrics: formulations containing 1 – 3 wt% coated nanoparticles and PE-g-MA maintained higher $\sigma_{(b)}$ and $\epsilon_{(b)}$ over exposure time than both neat PE and uncoated-NP composites. The correlation between (lower) CI and (higher) mechanical retention underscores the mechanistic link between photo-oxidation and embrittlement, and validates the UVA-340 exposure protocol used here as a discriminating screen for photostability differences [13]. Under UVA-340 accelerated weathering (0 – 500 h), polyethylene's FTIR carbonyl index (A1715/A1463) rises monotonically for all formulations: neat PE increases fastest, 3 wt% uncoated oxide nanoparticles show an intermediate rate, and 2 wt% coated nanoparticles with PE-g-MA the slowest. Lower slopes

indicate suppressed photo-oxidation via UV attenuation and passivated filler–matrix interfaces (Figure 2).

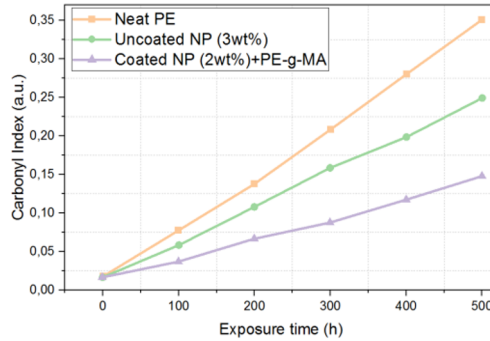


Fig. 2. Carbonyl Index versus $UVA - 340$ exposure: neat PE increases fastest; uncoated NP intermediate; coated NP + PE – g – MA shows minimal growth rate.

Thermal stability. Thermogravimetric analysis showed an upward shift of the onset degradation temperature (T_{onset}) for nano-reinforced PE compared with the matrix, with the largest gains for coated particles at 1 – 3 wt%. Oxidative induction time (OIT, DSC under O_2) increased in parallel, indicating a delayed onset of oxidative reactions-consistent with UV attenuation and radical quenching at (passivated) inorganic interfaces [14]. Differential scanning calorimetry further revealed modest increases in crystallinity (X_c) and, in some cases, slightly higher crystallization temperatures, pointing to a heterogeneous nucleation effect of the nanoparticles. The concomitant rise in storage modulus (E' , DMA) reflects both increased crystallinity and constrained amorphous dynamics near well-bound filler surfaces.

Mechanical properties. At ambient conditions, Young’s modulus (E) increased with nanoparticle loading, while yield strength (σ_y) showed a slight rise or plateau depending on dispersion quality. As often observed in particulate-filled semicrystalline polymers, elongation at break (ϵ_b) tended to decrease at higher loadings due to reduced ductility; however, the loss was minimized in the presence of PE – g – MA and with coated particles.

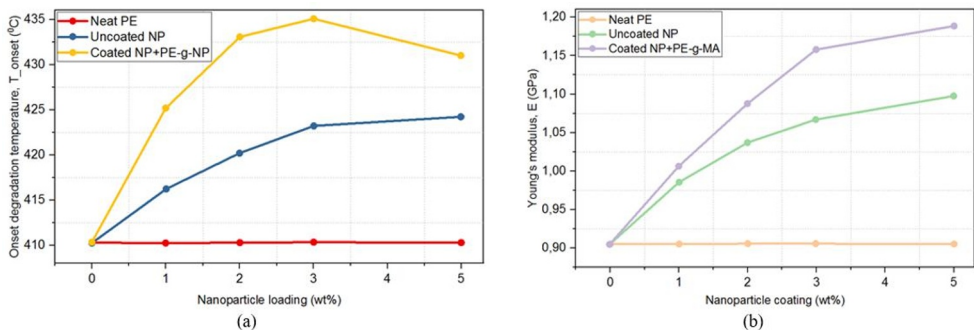


Fig. 3. (a): T_{onset} rises with loading; coated $TiO_2/ZnO + PE - g - MA$ peaks at 1 – 3 wt%. **(b):** Young’s modulus maximizes; gains diminish beyond 3 wt% from nanoparticle agglomeration.

A practical optimum emerged in the 1 – 3 wt% window, where stiffness gains and post-weathering retention were obtained with acceptable toughness trade-offs. Above ~3 – 5 wt%, agglomeration-driven stress concentrations and processing-induced defects outweighed benefits, eroding ϵ_b and long-term durability. Onset degradation temperature

(T_{onset}) versus nanoparticle loading. Coated $TiO_2/ZnO + PE - g - MA$ yields the largest T_{onset} increase, peaking at 1 – 3 wt%, while uncoated nanoparticles give moderate gains and neat PE remains unchanged (Figure 3a). Young's modulus versus loading shows trends, stiffness maximized at 1 – 3 wt% and diminishing returns beyond 3 wt% (Figure 3b).

Mechanistic picture. The observed performance trends can be rationalized by three synergistic mechanisms. First, optical screening: TiO_2/ZnO attenuate incident UV through absorption and scattering, lowering photon flux into the polymer and thereby reducing chromophore activation and radical initiation. Second, interfacial reinforcement: compatibilizer-mediated bonding ($PE - g - M$) improves stress transfer and limits cavitation/debonding under load, enhancing both initial strength and its retention after weathering. Third, morphological control: nanoparticles act as heterogeneous nucleants that modestly raise X_c and organize the semicrystalline morphology, increasing modulus and slowing oxygen diffusion in the interlamellar amorphous phase. Crucially, inorganic/organic passivation layers (SiO_2/Al_2O_3 shells, silanes, stearate) mitigate the well-known photocatalytic activity of bare oxides, which otherwise can accelerate chain scission and discoloration. The convergence of lower CI/YI growth rates, higher T_{onset}/OIT , and superior σ_b and ϵ_b retention under standardized UVA – 340 cycles supports this unified picture and delineates a robust processing–structure–property window for polyethylene nanocomposites optimized for outdoor or packaging applications [15].

Overall, coated TiO_2/ZnO at 1 – 3 wt% with $PE - g - MA$ compatibilization provide a balanced enhancement of UV durability, thermal stability, and mechanical performance. The limiting factors remain dispersion at higher loadings and viscosity growth during melt processing; these can be addressed by tailored surface chemistry and optimized compounding histories. Future work can map dose–response surfaces across particle size, shell thickness, and compatibilizer content, and couple accelerated weathering with multiscale modeling to refine lifetime predictions.

4 Conclusion

This study demonstrates that engineering polyethylene nanocomposites with TiO_2 and ZnO nanoparticles can simultaneously improve UV durability, thermal stability, and mechanical performance when surface chemistry and dispersion are rigorously controlled. Across melt-compounded and solution-cast routes, coated oxides combined with PE-g-MA produced the most robust responses. Relative to neat PE and uncoated fillers, 1–3 wt% coated nanoparticles increased T_{onset} and OIT, modestly raised crystallinity and T_c , and delivered higher storage and tensile moduli while maintaining superior retention of σ_b and ϵ_b during UVA-340 exposure. These gains arise from complementary mechanisms: optical attenuation of damaging radiation, compatibilizer-enabled interfacial stress transfer, and heterogeneous nucleation that organizes the semicrystalline morphology and retards oxidative transport. Practical limits emerge beyond ~3–5 wt% because agglomeration elevates viscosity, introduces defects, and compromises ductility. The results define a quantitative processing–structure–property window for packaging and outdoor components, and highlight the importance of oxide passivation layers to suppress photocatalysis. Future work should map design spaces involving particle size, shell thickness, and compatibilizer level; integrate organic stabilizers for synergistic protection; and couple accelerated weathering with field trials and multiscale modeling to refine lifetime predictions. Finally, scale-up studies and life-cycle assessments are encouraged to translate these formulations to durable, recyclable products globally.

References

1. Askarov M.N, Avezov I.Y, Bahranova U.I, Yuldosheva N.Yu, Ahmadov Kh.S. Analysis of Porous Water-Absorbing Materials Used in Solar Water Desalination. Sixteen International Conference on Thermal Engineering: Theory and Applications. June 18-20, Bucharest, Romania, (2025).
<https://journals.library.torontomu.ca/index.php/ictea/article/view/2526>.
2. I.Kh. Tychiev, Kh.S. Ahmadov, A.Yu. Ibodullaev, K.Yu. Rashidov, S.A Boltaev. Analysis of scientific publications based on the integration of polymer composite materials into solar thermal devices. Sixteen International Conference on Thermal Engineering: Theory and Applications. June 18-20, Bucharest, Romania, (2025).
<https://journals.library.torontomu.ca/index.php/ictea/article/view/2533>.
3. T. Tsuzuki, R. He, J. Wang, L. Sun, X. Wang, and R. Hocking, Reduction of the photocatalytic activity of ZnO nanoparticles for UV protection applications, *Int. J. Nanotechnol.*, **9**, no. 10–12, pp. 1017–1029, (2012).
4. G. Ascensão, G. Beersaerts, M. Marchi, M. Segata, F. Faleschini, and Y. Pontikes, Shrinkage and mitigation strategies to improve the dimensional stability of CaO-FeOx-Al₂O₃-SiO₂ inorganic polymers, *Materials*, **2**, no. 22, pp. 3679, (2019).
5. M. Anis Sofiah, O. H. Lin, H. M. Akil, and A. Villagracia, Effect of compatibiliser on the accelerated weathering performance of polypropylene–silica nanocomposites, *Mater. Res. Innov.*, **18**, no. sup6, pp. S6-433, (2014).
6. D. Yang, A. G. Ramu, and D. Choi, Synthesis of transparent ZnO–TiO₂ and its nanocomposites for ultraviolet protection of a polyethylene terephthalate (PET) film, *Catalysts*, **2**, no. 12, p. 1590, (2022).
7. S. Clarizio and R. Tatara, Tensile strength, elongation, hardness, and tensile and flexural moduli of injection-molded TPS filled with glycerol-plasticized DDGS, *J. Polym. Environ.*, **21**, no. 3, pp. 623–630, (2013).
8. I. Ozsoy, A. Demirkol, A. Mimaroglu, H. Unal, and Z. Demir, The influence of micro- and nano-filler content on the mechanical properties of epoxy composites, *Strojniski Vestnik Journal Mech. Eng.*, **61**, no. 10, (2015).
9. S. V. Kumar, K. S. Kumar, H. S. Jailani, and G. Rajamurugan, “Mechanical, DMA and sound acoustic behaviour of flax woven fabric reinforced epoxy composites,” *Mater. Res. Express*, **7**, no. 8, pp. 085302, (2020).
10. A. L. Andradý et al., Effects of UV radiation on natural and synthetic materials, *Photochem. Photobiol. Sci.*, **22**, no. 5, pp. 1177–1202, (2023).
11. W.-S. Cho, B.-C. Kang, J. K. Lee, J. Jeong, J.-H. Che, and S. H. Seok, Comparative absorption, distribution, and excretion of titanium dioxide and zinc oxide nanoparticles after repeated oral administration, *Part. Fibre Toxicol.*, **10**, no. 1, pp. 9, (2013)
12. H. M. Pires, L. C. Mendes, S. P. Cestari, and V. J. R. R. Pita, Effect of weathering and accelerated photoaging on PET/PC (80/20 wt/wt%) melt extruded blend, *Mater. Res.*, **18**, no. 4, pp. 763–768, (2015).
13. E. Gauthier, B. Laycock, F.-M. Cuoq, P. Halley, and K. George, Correlation between chain microstructural changes and embrittlement of LLDPE-based films during photo- and thermo-oxidative degradation, *Polym. Degrad. Stab.*, **98**, no. 1, pp. 425–435, (2013).
14. S. E. Lapuk, M. A. Ziganshin, R. A. Larionov, T. A. Mukhametzhanov, C. Schick, and A. V. Gerasimov, Differential scanning calorimetry investigation of crystallization kinetics and glass-forming ability of sulfonamides, *J. Non-Cryst. Solids*, **600**, pp. 122038, (2023).

15. S. Chupradit et al., “Morphological control: properties and applications of metal nanostructures,” *Adv. Mater. Sci. Eng.*, **2022**, no. 1, pp. 1971891, (2022).